

Yufei Chen

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OVERVIEW

My research focuses on the intersection of cybersecurity and artificial intelligence (AI). Currently, I am exploring new attack surfaces introduced by AI components and investigating how AI can be utilized to enhance existing security measures.

My (co-)first-author papers have been published in top security or machine learning venues: **USENIX Security** ($\times 2$), **NDSS** ($\times 1$), **TDSC** ($\times 1$), and **NeurIPS** ($\times 1$). For the latest updates on my publications, please visit [my Google Scholar profile](#) (**Google scholar citations: 1400+**).

EDUCATION

- 09/2019 – 06/2024 **City University of Hong Kong**, Hong Kong SAR, China
Ph.D. in Computer Science (Dual Ph.D. Programme)
Supervisor: Cong Wang
- 09/2016 – 12/2023 **Xi'an Jiaotong University**, Xi'an, China
Ph.D. in Control Science and Engineering (Dual Ph.D. Programme)
Supervisor: Chao Shen
- 02/2014 – 06/2016 **Xi'an Jiaotong University**, Xi'an, China
B.B.A. in Business Administration
- 09/2012 – 06/2016 **Xi'an Jiaotong University**, Xi'an, China
B.Eng. in Electrical Engineering

WORKING EXPERIENCES

- 01/2026 – PRESENT **City University of Hong Kong**, Hong Kong SAR, China
Research Assistant Professor
Department of Computer Science
- 06/2024 – 01/2026 **City University of Hong Kong**, Hong Kong SAR, China
Postdoc
Department of Computer Science
Supervisor: Cong Wang
- 03/2018 – 11/2018 **Qihoo 360**, Beijing, China
Research Intern
Research Center for Intelligence Security
Supervisor: Kang Li

PUBLICATIONS

(*: co-first authors)

CONFERENCE
PAPERS

- C15. Longxiang Wang*, Lei Xu*, **Yufei Chen**, Ying Zou, and Cong Wang. “ALERT: Machine Learning-Enhanced Risk Estimation for Databases Supporting Encrypted Queries,” in *Proceedings of 34th USENIX Security Symposium (USENIX Security)*, 2025. (Help with research idea refinement and paper editing.)
- C14. Sen Peng, **Yufei Chen**, Cong Wang, and Xiaohua Jia. “Intellectual Property Protection of Diffusion Models via the Watermark Diffusion Process,” in *Proceedings of 25th International Web Information Systems Engineering Conference (WISE)*, 2024. (Help with research idea refinement and paper editing.)
- C13. Xuanqi Gao, Juan Zhai, Shiqing Ma, Chao Shen, **Yufei Chen**, and Shiwei Wang. “CILATE: Towards Fairer Class-based Incremental Learning by Dataset and Training Refinement,” in *Proceedings of 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2023. (Help with research idea refinement and paper editing.)
- C12. **Yufei Chen**, Chao Shen, Yun Shen, Cong Wang, and Yang Zhang. “Amplifying Membership Exposure via Data Poisoning,” in *Proceedings of 36th Conference on Neural Information Processing Systems (NeurIPS)*, 2022. (Lead the research project, from algorithm design, implementation, and paper writing.)
- C11. **Yufei Chen**, Chao Shen, Cong Wang, and Yang Zhang. “Teacher Model Fingerprinting Attacks Against Transfer Learning,” in *Proceedings of 31st USENIX Security Symposium (USENIX Security)*, 2022. (Lead the research project, from algorithm design, implementation, and paper writing.)
- C10. Xuanqi Gao, Juan Zhai, Shiqing Ma, Chao Shen, **Yufei Chen**, and Qian Wang. “FairNeuron: Improving Deep Neural Network Fairness with Adversary Games on Selective Neurons,” in *Proceedings of 44th International Conference on Software Engineering (ICSE)*, 2022. (Help with research idea refinement and paper editing.)
- C9. Junhao Zhou*, **Yufei Chen***, Chao Shen, and Yang Zhang. “Property Inference Attacks Against GANs,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2022. (Lead evaluation design and paper writing.)
- C8. Qixue Xiao*, **Yufei Chen***, Chao Shen, Yu Chen, and Kang Li. “Seeing is Not Believing: Camouflage Attacks on Image Scaling Algorithms,” in *Proceedings of 28th USENIX Security Symposium (USENIX Security)*, 2019. (Lead attack algorithm design, implementation, and paper writing.)

- C7. Chao Shen, Qi Lv, Zhao Wang, **Yufei Chen**, and Xiaohong Guan. “Hand-Interactive Behavior Analysis for User Authentication Systems with Wrist-Worn Devices,” in *Proceedings of 2018 International Conference on Information, Cybernetics, and Computational Social Systems (ICCSS)*, 2018. Best Paper Award 🏆
- C6. Xiaozhi Liu, Chao Shen, and **Yufei Chen**. “Multi-source Interactive Behavior Analysis for Continuous User Authentication on Smartphones,” in *Proceedings of 2018 Chinese Conference on Biometric Recognition (CCBR)*, 2018.
- C5. Zhao Wang, Chao Shen, and **Yufei Chen**. “Handwaving Authentication: Unlocking Your Smartwatch Through Handwaving Biometrics,” in *Proceedings of 2017 Chinese Conference on Biometric Recognition (CCBR)*, 2017.
- C4. Chao Shen, Ziqiang Ren, **Yufei Chen**, and Zhao Wang. “On Using Wearable Devices to Steal Your Passwords: A Fuzzy Inference Approach,” in *Proceedings of 2017 International Symposium on Cyberspace Safety and Security*, 2017.
- C3. Zhanpei Jia, Chao Shen, Xiao Yi, **Yufei Chen**, Tianwen Yu, and Xiaohong Guan. “Big-data analysis of multi-source logs for anomaly detection on network-based system,” in *Proceedings of 13th IEEE Conference on Automation Science and Engineering (CASE)*, 2017.
- C2. **Yufei Chen**, Chao Shen, Zhao Wang, and Tianwen Yu. “Modeling Interactive Sensor-Behavior with Smartphones for Implicit and Active User Authentication,” in *Proceedings of 2017 IEEE International Conference on Identity, Security and Behavior Analysis (ISBA)*, 2017.
- C1. Chao Shen, **Yufei Chen**, and Gengshan Yang. “On Motion-Sensor Behavior Analysis for Human-Activity Recognition via Smartphones,” in *Proceedings of 2016 IEEE International Conference on Identity, Security and Behavior Analysis (ISBA)*, 2016.
- J11. Sen Peng, **Yufei Chen**, Jie Xu, Zizhuo Chen, Cong Wang, and Xiaohua Jia. “Intellectual Property Protection of DNN Models,” *World Wide Web*, vol. 26, pp. 1877-1911, 2023.
- J10. Kaidi Jin, Tianwei Zhang, Chao Shen, **Yufei Chen**, Ming Fan, Chenhao Lin, and Ting Liu. “Can We Mitigate Backdoor Attack Using Adversarial Detection Methods?,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, vol. 20, no. 4, pp. 2867-2881, 2023.
- J9. Xinyan Li, **Yufei Chen**, Cong Wang, and Chao Shen. “When Deep Learning Meets Differential Privacy: Privacy, Security, and More,” *IEEE Network*, vol. 35, no. 6, pp. 148-155, 2021.
- J8. **Yufei Chen**, Chao Shen, Cong Wang, Qixue Xiao, Kang Li, and Yu Chen. “Scaling Camouflage: Content Disguising Attack Against Computer Vision Applications,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, vol. 18, no. 5, pp. 2017-2028, 2021.

- J7. Chao Shen, Zhao Wang, Chengxiang Si, **Yufei Chen**, and Xiaojie Su. “Waving Gesture Analysis for User Authentication in the Mobile Environment,” *IEEE Network*, vol. 34, no. 2, pp. 57-63, 2020.
- J6. **Yufei Chen**, Chao Shen, Qian Wang, Qi Li, Cong Wang, Shouling Ji, Kang Li, and Xiaohong Guan. “Security and Privacy Risks in Artificial Intelligence Systems (人工智能系统安全与隐私风险),” *Journal of Computer Research and Development*, vol. 56, no. 10, pp. 2135-2150, 2019. (in Chinese)
- J5. Chao Shen, **Yufei Chen**, Yao Liu, and Xiaohong Guan. “Adaptive Human-Machine Interactive Behavior Analysis With Wrist-Worn Devices for Password Inference,” *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 29, no. 12, pp. 6292-6302, 2018.
- J4. Chao Shen, **Yufei Chen**, and Xiaohong Guan. “Performance Evaluation of Implicit Smartphones Authentication via Sensor-Behavior Analysis,” *Information Sciences*, vol. 430-431, pp. 538-553, 2018.
- J3. Chao Shen, **Yufei Chen**, Xiaohong Guan, and Roy Maxion. “Pattern-Growth based Mining Mouse-Interaction Behavior for an Active User Authentication System,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, vol. 17, no. 2, pp. 335-349, 2018.
- J2. Chao Shen, **Yufei Chen**, Gengshan Yang, and Xiaohong Guan. “Toward Hand-Dominated Activity Recognition Systems With Wristband-Interaction Behavior Analysis,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 50, no. 7, pp. 2501-2511, 2018.
- J1. Chao Shen, Yuanxun Li, **Yufei Chen**, Xiaohong Guan, and Roy Maxion. “Performance Analysis of Multi-Motion Sensor Behavior for Active Smartphone Authentication,” *IEEE Transactions on Information Forensics and Security (TIFS)*, vol. 13, no. 1, 2018.

TEACHING

2026 Spring	CS5293 (Topics on Information Security), PG, Tutor , CityUHK(DG)
2025 Fall	CS6535 (Guided Study in Artificial Intelligence), PG, Tutor , CityUHK(DG)
2025 Spring	CS6290 (Privacy-enhancing Technologies), PG, Tutor , CityUHK(DG)
2025 Spring	CS2310 (Computer Programming), UG, Tutor , CityUHK
2024 Fall	CS2311 (Computer Programming), UG, Tutor , CityUHK

AWARDS AND HONORS

11/2025	NeurIPS 2025 Top Reviewer (Top 8%)
11/2024	NeurIPS 2024 Top Reviewer (Top 10%)
07/2022	USENIX Security 2022 Student Grant
11/2021	Outstanding Presenter Award of the 14th USTC PhD Student Workshop
03/2021	The Shaanxi Provincial Natural Science Award (First Class, 4/4)
12/2020	National Scholarship for Ph.D. Student (Top 0.2%)

06/2020	Outstanding Graduate Student of XJTU
10/2019	Cyber Security Scholarship (awarded annually to 100 graduate students nationwide)
12/2018	National Scholarship for Ph.D. Student (Top 0.2%)
08/2018	IEEE ICCSS Best Paper Award
12/2017	National Scholarship for Graduate Student (Top 0.2%)
12/2015	National Scholarship for Undergraduate Student (Top 0.2%)
12/2013	National Endeavor Scholarship (Top 3%)
2012 – 2016	Outstanding Undergraduate Student of XJTU

SERVICES

PC MEMBER REVIEWER	NeurIPS 2025, 2024 (Top reviewer $\times 2$ 🏆) ICLR 2026, 2025 ICML 2025 AAAI 2025 AISTATS 2026, 2025 LAMPS-CCS 2025 MSN 2025, 2024 SecTL-AsiaCCS 2025, 2024, 2023
EXTERNAL REVIEWER	USENIX Security 2024, 2023 NDSS 2022 ACM CCS 2023 ESORICS 2024 RAID 2022, 2021 ACSAC 2022 ACNS 2022 AsiaCCS 2021 ICICS 2021
JOURNAL REVIEWER	IEEE Transactions on Dependable and Secure Computing IEEE Transactions on Information Forensics and Security IEEE Transactions on Mobile Computing ACM Transactions on Privacy and Security IEEE Transactions on Neural Networks and Learning Systems IEEE Internet of Things Journal IEEE Transactions on Cybernetics ACM Transactions on Sensor Networks Transactions on Machine Learning Research IEEE Transactions on Automation Science and Engineering Neural Networks Computers & Security Information Sciences